AP PRECALCULUS

Unit 1 • Lesson 1.9 notes

Lesson 1.9 • Rational Functions and Vertical Asymptotes

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Locate vertical asymptotes of a rational function.
- Describe the behavior near a vertical asymptote with one-sided limits.

Vocabulary. vertical asymptote, one-sided limit, $\lim_{x\to a^-}$, $\lim_{x\to a^+}$, multiplicity.

1. Where are the vertical asymptotes?

Key idea

After factoring and cancelling common factors, each remaining zero a of the denominator gives a vertical asymptote $x = a$. Near it, $|r(x)|$ grows without bound.

2. Which way does the graph go?

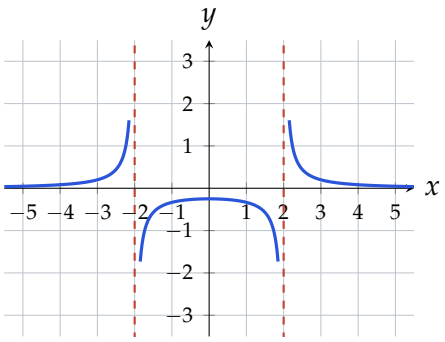
Look at the sign of each factor just left and just right of a. If $(x-a)$ appears an odd number of times in the denominator, the sign of r flips across the asymptote; an even number of times, it stays the same.

Example 1

Describe $r(x) = \dfrac{x+1}{(x-2)^2}$ near $x = 2$.

Solution. Near 2 the numerator is close to $3 > 0$ and $(x-2)^2 > 0$ on both sides. So $\lim_{x\to 2^-} r(x) = +\infty$ and $\lim_{x\to 2^+} r(x) = +\infty$.

Example 2 • Reading a graph



$r(x) = \dfrac{1}{(x-2)(x+2)}$.

Solution. $\lim_{x\to -2^-} r(x) = +\infty$, $\lim_{x\to -2^+} r(x) = -\infty$, $\lim_{x\to 2^-} r(x) = -\infty$, $\lim_{x\to 2^+} r(x) = +\infty$.

Common mistakes

- Putting a vertical asymptote at a value where the factor cancels (that is a hole, Lesson 1.10).
- Guessing the direction of the infinite limit instead of testing signs.

Quick check

1. Vertical asymptotes of $\dfrac{x-3}{x^2-9}$? 2. $\lim_{x\to 0^-} \dfrac{2}{x^3} = ?$

Answers. 1. Only $x = -3$ ($x = 3$ cancels). 2. $-\infty$.